AKASH YADAV

Houston, TX ⋄ ayadav4@uh.edu ⋄ akashyadav0210.github.io ⋄ Google Scholar

RESEARCH INTERESTS

Uncertainty-aware Scientific Machine Learning, Scientific Foundation Models, Uncertainty Quantification and Model Calibration, Computational Mechanics, and Bayesian Optimization

EDUCATION

Doctor of Philosophy, Civil Engineering (GPA: 4.0/4.0) *08/2023 - 05/2027 (expected)*
Dissertation: Quantifying and Reducing Model Error via Stochastic Representations
University of Houston (UH) - Houston, USA

Master of Technology (Research), Civil Engineering (GPA: 8.3/10) *10/2020 - 06/2023*
Thesis: SHM Accounting for Thermal Variability and Damage using ABC
Indian Institute of Science (IISc) - Bangalore, India

Bachelor of Technology, Civil Engineering (First Division, GPA: 8.75/10) *07/2014 - 05/2018*
Indian Institute of Technology (IIT) - Roorkee, India

SKILLS

ML/UQ	Transformers, LoRA, PINNs, CNNs, GANs, Gaussian Processes, BO, MCMC
Programming	Python, PyTorch, GPyTorch, BoTorch, NumPy, SciPy, MATLAB, Julia
HPC/CAE	Distributed training, SLURM, Git, Abaqus, LS-DYNA, FEniCS, Gmsh

RESEARCH EXPERIENCE

Graduate Research Assistant - UH (Advisor: Dr. Ruda Zhang) *08/2023 - Present*

- **Uncertainty-aware foundation models:** Developed an inference-time stochastic attention mechanism that equips pretrained scientific foundation models (SciFMs) with calibrated predictive uncertainty without retraining the backbone. Evaluated the method on mechanics-focused SciFMs, yielding sharp, well-calibrated predictions while preserving predictive accuracy.
- **Model error correction:** Developed a stochastic reduced-order model (SROM) framework for model error correction in computational mechanics, improving predictive uncertainty under limited data.
- **Bayesian optimization under uncertainty:** Built a Bayesian optimization under uncertainty workflow to tune hyperparameters in stochastic models, reducing data requirements and accelerating tuning by 40× relative to scalar bounded optimization and 15× relative to conventional Gaussian-process Bayesian optimization baselines.
- **Stochastic subspaces for model error:** Proposed stochastic subspace models via probabilistic principal component analysis and the Bootstrap for model error characterization in computational mechanics.

Graduate Research Assistant - IISc (Advisor: Dr. Ananth Ramaswamy) *10/2020 - 06/2023*

- Developed a method based on Approximate Bayesian Computation (ABC) for damage detection under varying temperature conditions, and extended it to incorporate damage-induced nonlinearity.

PUBLICATIONS

1. **Yadav, A.**, Adebiyi, T.A., & Zhang, R. (2026). "Calibrating Scientific Foundation Models with Inference-Time Stochastic Attention." *arXiv*. arXiv:2604.19530. Under Review.
2. **Yadav, A.** & Zhang, R. (2026). "Bayesian Optimization under Uncertainty for Training a Scale Parameter in Stochastic Models." *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering*, 12(4). doi:10.1061/AJRUA6.RUENG-1854.
3. **Yadav, A.** & Zhang, R. (2026). "Nonparametric Stochastic Subspaces via the Bootstrap Method for Characterizing Model Error." *ASCE-ASME Journal of Risk and Uncertainty in Engineering Systems, Part A: Civil Engineering*. doi:10.1061/AJRUA6.RUENG-1948.
4. **Yadav, A.** & Zhang, R. (2026). "Stochastic Subspace via Probabilistic Principal Component Analysis for Characterizing Model Error." *Computational Mechanics*, 77, 1141–1156. doi:10.1007/s00466-025-02701-6.

CONFERENCE PRESENTATIONS

1. **A. Yadav**, T. Adebiyi, & R. Zhang, Calibrated Uncertainty for Fine-tuned Scientific Foundation Models via Stochastic Attention, 17^{th} *World Congress on Computational Mechanics* & 10^{th} *European Congress on Computational Methods in Applied Sciences and Engineering (WCCM/ECCOMAS)*, Munich, Germany, July 19-24, 2026.
2. **A. Yadav** & R. Zhang, Stochastic reduced-order modeling for model error characterization and correction, 18^{th} *US National Congress on Computational Mechanics (USNCCM)*, Chicago, USA, July 20-24, 2025.
3. **A. Yadav** & R. Zhang, Stochastic subspace via bootstrap for model-form uncertainty, *Conference on Applied AI & Scientific Machine Learning (CASML)*, IISc, Bangalore, India, December 14-18, 2024.
4. **A. Yadav** & R. Zhang, Stochastic subspace via probabilistic principal component analysis for model-form uncertainty, 16^{th} *World Congress on Computational Mechanics* & 4^{th} *Pan American Congress on Computational Mechanics (WCCM/PANACM)*, Vancouver, Canada, July 21-26, 2024.
5. **A. Yadav** & A. Ramaswamy, Structural health monitoring of steel truss bridges subjected to environmental variability, 8^{th} *International Congress on Computational Mechanics & Simulation (ICCMS)*, IIT, Indore, India, December 9-11, 2022.

HONORS AND AWARDS

Chevron Energy Graduate Fellows Award, UH	*2026 - 2027*
Jimmie A. Schindewolf Academic Scholarship, UH	*08/2024 - 05/2025, 08/2026 - 12/2026*
Student Travel Award for WCCM/ECCOMAS 2026	*03/2026*
Future Faculty Program, UH	*08/2024 - 05/2025*
Presidential Fellowship, UH	*08/2023 - 05/2025*
Finalist, UQ-TTA Student Paper Competition, WCCM/PANACM	*07/2024*
Cullen Fellowship Travel Grant, UH	*05/2024*
Top 0.35% in IIT-JEE (out of 1.4 million candidates)	*06/2014*

TEACHING EXPERIENCE

Teaching Assistant, Foundation Engineering (CIVE 4369), UH	*01/2026 - 05/2026*
Reciter, Mechanics-I Statics (ENGR 2301), UH	*08/2024 - 12/2024*
Teaching Assistant, Mechanics-I Statics (ENGR 2301), UH	*08/2023 - 12/2023*

WORK EXPERIENCE

Senior Project Engineer - Indian Oil Corporation Limited *07/2018 - 09/2020*
Oversaw civil execution for an energy-efficient green building and process-unit expansion; coordinated contractors, ensured code compliance, and managed schedules across multiple work fronts.

Industrial Internship - RITES Limited *Summer 2017, 2016*
Designed highway bridge components, including superstructures and substructures, and performed finite element analysis of box culverts using Midas Civil. Designed retaining walls using both working stress and limit state methods, and applied IRC codes for culvert design using STAAD.Pro.

RELEVANT COURSEWORK

Mathematics for AI, Deep Learning, Learning with Data, Data-Driven Engineering, Bayesian Statistics, Optimization Methods, Numerical Methods, Finite Element Method, Structural System Identification.

EXTRACURRICULAR ACTIVITIES

Secretary/Joint Secretary, Taekwondo, Institute Sports Council, IIT Roorkee *2016 - 2018*
Organized training for 50+ students and coordinated participation in state/national competitions.

First Dan Black Belt in Taekwondo by World Taekwondo Federation	*08/2017*
Represented State Uttarakhand in **National Taekwondo Championship**	*12/2017*
Organizer and Instructor, Self Defense Camp, Unnat Bharat Abhiyaan	*03/2017*

ACADEMIC AND PROFESSIONAL AFFILIATIONS

United States Association for Computational Mechanics (USACM), Technical Thrust Area in Uncertainty Quantification and Probabilistic Modeling, Graduate Student Member.
Society for Industrial and Applied Mathematics (SIAM), Graduate Student Member.